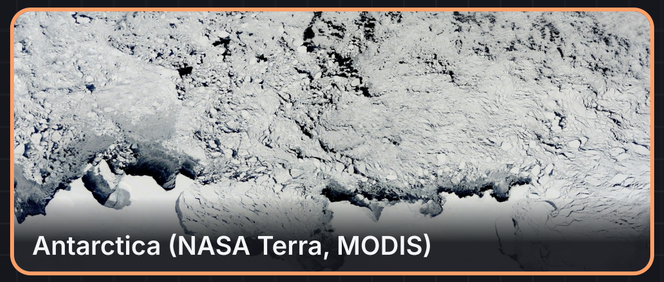
Pi & Planck

NOBEL PRIZE

The 2026 Nobel Prize in Physics

A cubic kilometre of Antarctic ice became a telescope for cosmic neutrinos.

THE LAUREATE



Antarctica (NASA Terra, MODIS)

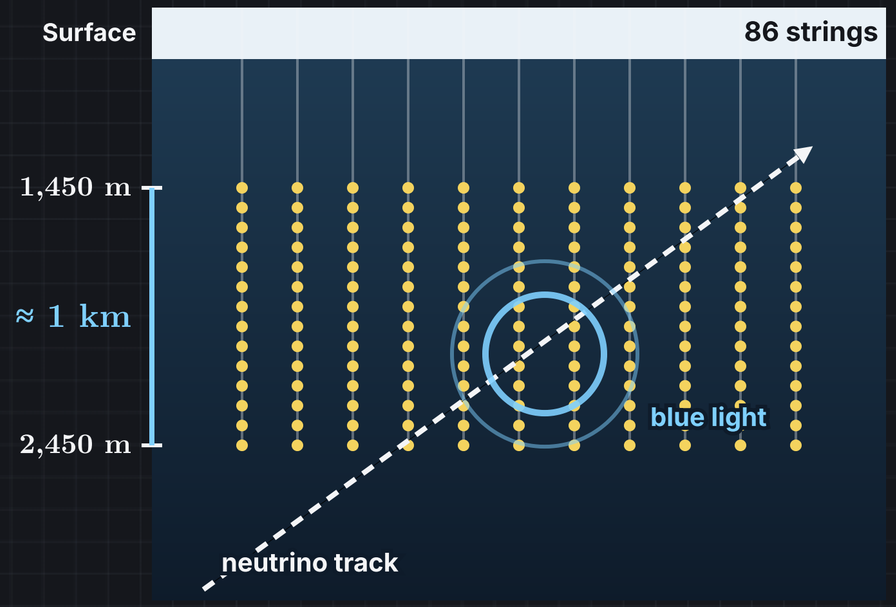
Francis Halzen

University of Wisconsin–Madison, USA

Born in Tienen, Belgium, 1944. Sole laureate.

For decisive contributions to the IceCube Neutrino Observatory and the discovery of high-energy neutrinos of cosmic origin.

HOW ICECUBE SEES



● 60 sensors on each

DOM = digital optical module · schematic, not to scale

FOUR DISCOVERIES

2013 — 28 cosmic neutrinos
First evidence of high-energy neutrinos from beyond the Solar System: 28 events, from about 30 TeV to 1.2 PeV.

2017 — A blazar named
A neutrino of about 290 TeV arrived from the direction of the flaring blazar TXS 0506+056.

2018 — An earlier flare
IceCube data from 2014–2015 showed an excess from the same direction (3.5σ).

2023 — The Milky Way glows
Neutrinos from the Galactic plane, found at 4.5σ with machine learning.

1 km³ of ice — Instrumented ice, from 1,450 m to 2,450 m below the surface.

5,160 sensors — Digital optical modules on 86 strings: 86 × 60.

6×10^{10} per cm² per s — Solar neutrinos reaching Earth, at far lower energy than IceCube's cosmic ones.

Sources: Nobel Prize press release and scientific background, nobelprize.org · IceCube, JINST 12 P03012 (2017) · Science 342, 1242856 (2013) · Science 361, eaat1378 and 147–151 (2018) · Science 380, 1338 (2023) · Borexino, Nature 587, 577 (2020) · Photo: NASA GSFC

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